Responsible AI Roadmap – D1, D2 and D3

Project "Integrating and Engineering AI Applications"

EC5, Responsible Engineering of AI Applications

Group: 2

Deliverable 1 – Use Cases

1. Perimeter & Scope

Goal of the AI application
The system aims to automate the intralogistics functions of a manufacturing workshop (transport of materials, parts and products between reception, storage, production and shipping areas) using a fleet of autonomous mobile robots. Each robot perceives its environment, reasons over a shared knowledge graph, plans its actions and coordinates with the other robots and with the machines (conveyors, robotic arms). Human operators delegate goals to the robots by voice and keep control over their behaviour. The objective is to demonstrate the technical feasibility of this automation, while studying its societal and environmental impacts and validating responsible engineering choices (regulation, sustainability, risk management).

Context of the AI Application
Project. Demonstrator built on the MAQ'IT platform (Laboratoire des transitions, Institut Henri Fayol, Mines Saint-Étienne), which represents the workshop of a generic manufacturing company. It covers the multi-robot hardware and software building block (mobile robots, robotic arms, conveyors) within a cyber-physical and social system (CPS). The robots, arms and conveyors are those of the platform. Expected benefits: less manual handling and walking, hence lower exposure to musculoskeletal disorders, which account for over 80% of recognised occupational diseases in France, logistics being among the most affected sectors. **AI technologies.** Environment perception (camera, lidar), knowledge graph and reasoning, single- and multi-agent action and trajectory planning, multi-agent coordination, voice interaction. The precise techniques have not been chosen yet. **Use.** Continuous human supervision: the human delegates goals, can change them dynamically and receives feedback from the robot ("yes", "no", "not understood", "possible", "impossible"). Robots and humans are the only mobile entities and share the same space, which raises safety issues. Out-of-scope uses to anticipate: using the robots to monitor staff, voice commands issued by an unauthorised person. Learning: perception and speech models are trained offline and frozen during operation, updated only through validated versions. Time-criticality: high for safety (avoidance and stopping are decided in real time, with no possible human validation), low for task allocation, which humans can review.

Domain. Industrial production and intralogistics. Experts to consult: project supervisors, MAQ'IT platform managers, logistics and production managers, HSE manager, experienced operators, staff representatives, Ethics and data regulator.

Data. Images and lidar point clouds of the workshop, voice commands, workshop map, knowledge graph rules. Data collected for the project [to be confirmed]. Identified risk: unintended capture of images and voices of people present.

Social. Benefits: less manual handling and fewer musculoskeletal disorders. Issues: substitution of human tasks by robots and its effect on jobs and skills; workload and stress when humans take over after a failure; safety of people sharing space with robots; inclusion of vulnerable employees (voice-only or sound-only interaction); privacy at work (cameras, lidar, microphones); human control over the robots; social dialogue with staff representatives.

Environmental. Energy consumed by the robot fleet (charging) and by the AI models (perception, speech, planning); material footprint of robots, batteries and electronics, their lifespan, repairability and end of life. Possible gains (optimised movements, fewer unnecessary trips) are to be assessed [hypothesis, to be quantified].

Regulatory. Frameworks to check [to be verified with the HSE manager, the DPO and up-to-date legal sources]: European regulation on AI systems (AI Act); personal data protection (GDPR) for voice and image data; machinery safety (European Machinery Regulation and standards for mobile robots and human-robot collaboration, for example ISO 3691-4 and ISO 10218); occupational health and safety and labour law (risk assessment, information and consultation of staff representatives); environmental rules on batteries and electronic waste.

2. Use Cases

Potential uses of the technology with involved stakeholders

Use case 1 : voice delegation of a transport task

Stakeholders: Operators, Employees with disabilities, Non-user employees present, External people, Managers, HSE manager, Staff representatives, IT and cybersecurity team, Company management, Ethics and data regulator

Reference Stakeholder: Operators

An operator asks a robot by voice to carry a part, a pallet or a batch of materials from one area of the workshop to another ("transport <item name>"). The robot answers "possible", "impossible" or "not understood", plans its route from the workshop map, then executes it. The operator can stop the robot at any time ("stop") or change the goal. On the way, robots meet people in corridors and shared areas: they detect them (camera, lidar), slow down or stop, and signal their presence or intention (LED, voice message). Sensitive situations include ambient noise, hearing-impaired or reduced-mobility people, and people not equipped like usual operators (visitors). Several robots also share the same corridors. The use case involves capturing images and sounds in the workshop. Failure and substitution cases are developed in use case 3.

Use case 2: production order for a product

Stakeholders: Operators, Quality control, Employees with disabilities, Non-user employees present, External people, Managers, HSE manager, Staff representatives, IT and cybersecurity team, Company management, Ethics and data regulator

Reference Stakeholder: Operators

The production manager delegates a high-level goal ("make <product name>"). The system plans the sequence of actions: collecting materials and components, bringing them to the relevant stations, then moving finished products to storage. When several robots need the same resource (conveyor, robotic arm, corridor), they distribute the goals and arbitrate access according to product urgency, machine utilisation and maintenance needs; the criteria must be understandable and contestable by the humans concerned. Humans working at the stations share space with the robots (detection, signals, vulnerable people). The human can follow progress, ask for explanations about the plan and modify it. Failure and substitution cases are developed in use case 3.

Use case 3: machine unavailability and human takeover

Stakeholders: Operators, Maintenance technicians, Quality control, Managers.

Reference Stakeholder: Maintenance technicians

A failure interrupts the robot task. The failure may come from a robot or a human (robot blocked, empty battery, misperception, wrong or ambiguous order, misunderstood command, machine or robot unavailable, operator absent, unavailable or overloaded). The system reports the failure, puts the robot in a safe state and reallocates the task: either a human

takes over the tasks (a human substitutes for the robot), or a robot takes over the tasks left by an absent or overloaded operator (a robot substitutes for the human). The human can accept or refuse the request. Maintenance technicians diagnose and repair the unavailable robot or machine, and quality control checks the products handled manually during the failure.

Deliverable 2 – Stakeholders

1. Stakeholder groups overview

We listed as many stakeholders as possible, grouped them following the typology of the course (affected stakeholders, intermediaries, AI Application Makers), and justified each one in relation to the use cases of Deliverable 1.

Key stakeholders (★) are those from whom we will draw an in-depth understanding of people, values and technology. The robots are the AI system itself and are not stakeholders.

Stakeholder group	Stakeholders
End-users	S01
Directly affected stakeholders	S02, S03, S04, S05, S06
Intermediaries (spokespersons)	S07, S08, S09
AI Application Makers (developers, facilitators)	S10, S11, S12

2. Persona of the involved stakeholders

★ S01 – Operators

Stakeholder group: End-user

Use case(s): UC1, UC2, UC3

Why a stakeholder: Give voice orders to the robots and work alongside them every day, share corridors and stations with them, and are the ones who take over a task, or are replaced by a robot, when something fails. They are the main stakeholder of the robots.

Actions and decisions w.r.t. the system: Delegate, modify or cancel goals by voice; stop a robot; accept or refuse a takeover request; report failures.

S02 – Maintenance technicians

Stakeholder group: Directly affected

Use case(s): UC3

Why a stakeholder: Maintain the robots and shared machines, whose failure triggers the substitution cases and whose availability drives the arbitration between robots; their repairs also determine the lifespan of the equipment.

Actions and decisions w.r.t. the system: Declare machines unavailable or back in service; diagnose faults; repair or replace parts.

S03 – Quality control

Stakeholder group: Directly affected

Use case(s): UC2, UC3

Why a stakeholder: Check that the products handled by the robots, including fragile or perishable ones, meet quality requirements, also when handling is done manually after a failure.

Actions and decisions w.r.t. the system: Define handling constraints per product; block or release a batch.

S04 – Employees with disabilities (reduced mobility, hearing or speech impairment)

Stakeholder group: Directly affected – Vulnerable group

Use case(s): UC1, UC2

Why a stakeholder: May be unable to use voice commands, to perceive the robots' signals or to move out of their way quickly.

Actions and decisions w.r.t. the system: Use alternative interaction means if available; depend on inclusive design choices.

S05 – Non-user employees present in the workshop (e.g. cleaning and facility staff)

Stakeholder group: Directly affected

Use case(s): UC1, UC2

Why a stakeholder: Work in the shared space without using the system and must be detected and respected by the robots, whose sensors also capture them.

Actions and decisions w.r.t. the system: Enter robot areas; may be slowed or blocked by robots; report unsafe situations.

S06 – External people (visitors, clients, suppliers, carriers)

Stakeholder group: Directly affected

Use case(s): UC1, UC2

Why a stakeholder: Visitors cross the workshop without training and are detected by the robots; suppliers and carriers deliver and collect in the areas the robots serve; clients receive products whose quality and delays depend on production flows.

Actions and decisions w.r.t. the system: Follow safety instructions; deliver and collect goods; no control over the system.

★ S07 – Managers

Stakeholder group: Intermediary – Spokesperson

Use case(s): UC1, UC2, UC3

Why a stakeholder: Speak on behalf of operators, assign and prioritise transport and production tasks, and decide how failures are handled and how tasks are shared between humans and robots.

Actions and decisions w.r.t. the system: Set delegation rules, priorities and arbitration criteria; launch production orders; decide on task reallocation and on deployment.

S08 – Health & safety (HSE) manager

Stakeholder group: Intermediary – Spokesperson

Use case(s): UC1, UC2

Why a stakeholder: Responsible for compliance with safety rules and for everyone's safety in the workshop: human-robot coexistence. Monitors health at work and work accidents, and can link the harms people suffer to the system's design (collisions, workload).

Actions and decisions w.r.t. the system: Set speed and stopping rules, safety zones and incident procedures; can halt the deployment. Advise on health risks; analyse accidents; recommend protective measures.

★ S09 – Staff representatives (works council, unions)

Stakeholder group: Intermediary – Spokesperson

Use case(s): UC1, UC2

Why a stakeholder: Defend employees' interests on jobs, substitution of human tasks by robots, surveillance, workload and consent across all uses of the system.

Actions and decisions w.r.t. the system: Are informed and consulted; can contest or negotiate the conditions of deployment.

S10 – IT and cybersecurity team

Stakeholder group: AI Application Maker – Developers

Use case(s): UC1, UC2

Why a stakeholder: Operates the network, data infrastructure and access rights, and protects the system against attacks and misuse.

Actions and decisions w.r.t. the system: Manage accounts, data storage and security; respond to incidents.

★ S11 – Company management

Stakeholder group: AI Application Maker – Facilitators

Use case(s): UC1, UC2

Why a stakeholder: Decides to invest in automation, sets its business objectives and budget, and is the employer legally responsible for the workplace.

Actions and decisions w.r.t. the system: Approve, fund or stop the project; set business, social and sustainability priorities.

S12 – Ethics and data regulator

Stakeholder group: AI Application Maker – Facilitators

Use case(s): UC1, UC2

Why a stakeholder: Ensures compliance with ethical, regulatory (GDPR), and AI standards (transparency, non-discrimination, human control, data hosting). Applies to UC1 (audio/video capture in the workshop) and UC2 (operator detection and plan explainability).

Actions and decisions w.r.t. the system: Defines legal and ethical requirements, audits the system, sets deployment constraints (sensor limitations, emergency stop, explainability), and manages incidents or non-compliance.

3. Comprehensive map of involved stakeholders with the corresponding use cases

X = involved in the use case; NA = not involved. Purple columns are key stakeholders (★). Codes refer to the persona boxes of section 2.

Use cases / Stakeholders	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	S11	S12
UC1 – Voice delegation of a transport task	X	NA	NA	X	X	X	X	X	X	X	X	X
UC2 – Production order for a product	X	NA	X	X	X	X	X	X	X	X	X	X
UC3 – Machine unavailability and human takeover	X	X	X	NA	NA	NA	X	NA	NA	NA	NA	NA

S01 Operators | S02 Maintenance technicians | S03 Quality control | S04 Employees with disabilities | S05 Non-user employees present | S06 External people | S07 Managers | S08 HSE manager | S09 Staff representatives | S10 IT and cybersecurity team | S11 Company management | S12 Ethics and data regulator

Deliverable 3 – Stakeholders × values, persona-value cards and list of values

Comprehensive map of stakeholders and values. Purple columns are key stakeholders (★).

Values / Stakeholders	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	S11	S12
SOCIAL theme												
Safety (people, technical safety)	X	X	NA	X	X	X	X	X	X	NA	X	NA
Human control	X	X	NA	X	NA	NA	NA	X	X	NA	NA	X
Ease of use	X	NA	NA	X	NA	NA	NA	NA	NA	NA	NA	NA
Reliability	X	X	X	NA	NA	X	X	NA	NA	NA	X	NA
Fairness (diversity, non-discrimination)	X	NA	NA	X	NA	NA	X	NA	X	NA	NA	X
Social well-being (workload, jobs)	X	NA	NA	X	NA	NA	NA	X	X	NA	NA	NA
ENVIRONMENTAL theme												
Environmental well-being	NA	NA	NA	NA	NA	NA	NA	X	NA	NA	X	NA
Maintainability (lifespan, repair)	NA	X	NA	NA	NA	NA	NA	NA	NA	X	NA	NA
REGULATORY theme												
Confidentiality	X	NA	NA	X	X	X	NA	X	X	X	NA	X
Explainability	NA	X	X	NA	X	NA	X	X	X	NA	NA	X
Accountability	NA	NA	X	NA	NA	NA	X	X	X	X	X	X
Deployment & sovereignty	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	X	X
Regulatory compliance	NA	NA	NA	NA	NA	NA	NA	X	NA	NA	X	X
Technical and economic values												
Robustness	NA	NA	NA	NA	NA	NA	X	X	NA	X	NA	NA
Economic sustainability (cost, productivity)	NA	NA	NA	NA	NA	NA	X	NA	NA	X	X	NA

S01 Operators | S02 Maintenance technicians | S03 Quality control | S04 Employees with disabilities | S05 Non-user employees present | S06 External people | S07 Managers | S08 HSE manager | S09 Staff representatives | S10 IT and cybersecurity team | S11 Company management | S12 Ethics and data regulator

The persona-value cards below are living documents: we complete them during our interviews, research and discussions, and every piece of information carries the source it comes from. Cards are made for the 4 key stakeholders (★): S01, S07, S09 and S11.

1. Values for stakeholder personas & use cases

Persona-value cards of the key stakeholders

★ S01 – Operators	
Group / use cases	End-user / UC1, UC1, UC3
Primary goal	Carry out their transport and production tasks on time, at the expected quality, safely and without physical strain.
Likes about their job and interaction with the company	Relief from heavy and repetitive carrying; reliable and timely supply of materials; being able to hand over a task simply, by voice; pride in skilled work.
Dislikes about their job and interaction with the company	Time pressure and being blamed for delays; not knowing who is responsible when a robot fails; robots getting in the way; being asked to take over tasks at short notice; feeling watched by cameras and microphones; repeating orders in a noisy workshop.
Company / group values	Safety, efficiency, reliability, quality, fair treatment between teams, team spirit.
Other stakeholders who support them	Managers, colleagues, quality control, maintenance technicians, development team (support and fixes).
How they use the technology to achieve their goal	Give voice orders, stop robots, change goals, receive materials and products delivered by the robots, share space with them, and accept or refuse to take over tasks when a machine or robot fails.
Policies, regulations, social norms, technical constraints	Workshop safety rules and protective equipment; working-time and labour law, including on task allocation; data protection rules for voice and image capture; commands must work in a noisy workshop; production quality procedures.
Stakeholder values (from the stakeholder × values matrix)	M: Safety (people, technical safety), Ease of use, Reliability, Human control, Confidentiality H: Social well-being (workload, jobs), Fairness (diversity, non-discrimination)
Societal values extracted from the persona	Decent work, health and safety, privacy, human agency, fairness.

★ S07 – Managers	
Group / use cases	Intermediary – Spokesperson, UC1, UC2, UC3
Primary goal	Ensure smooth, cost-effective flows and meet production targets in terms of quality, cost and deadlines.

Likes about their job and interaction with the company	Visibility on flows; predictable and high throughput; stable quality; clear priorities; a motivated team; innovation that works.
Dislikes about their job and interaction with the company	Breakdowns, blocked flows and line stoppages; staff resistance; unclear return on investment; conflicting priorities between teams; opaque automated decisions that cannot be explained to the teams; being held responsible for incidents involving automated equipment.
Company / group values	Efficiency, productivity, cost control, reliability, quality, safety, employee well-being.
Other stakeholders who support them	Operators, quality control, maintenance manager, IT team, development team, company management.
How they use the technology to achieve their goal	Authorise delegation rules, set transport and production priorities and arbitration criteria between robots, launch production orders, and decide on task reallocation and deployment when something fails.
Policies, regulations, social norms, technical constraints	Budget; production plans; quality standards; maintenance windows; labour law; safety rules; delivery commitments to suppliers and customers.
Stakeholder values (from the stakeholder × values matrix)	M: Safety (people, technical safety), Reliability, Robustness, Explainability H: Economic sustainability (cost, productivity), Accountability, Fairness (diversity, non-discrimination)
Societal values extracted from the persona	Economic sustainability, decent work, product quality and safety, accountability.

★ S09 – Staff representatives	
Group / use cases	Intermediary – Spokesperson / UC1, UC2
Primary goal	Protect employees' jobs, working conditions and rights, and be informed and consulted about changes.
Likes about their job and interaction with the company	Genuine social dialogue; transparency from management; early involvement in technology projects.
Dislikes about their job and interaction with the company	Surveillance through cameras and microphones; deployment decided without consultation; automation used to cut jobs without discussion; opaque algorithms.
Company / group values	Solidarity, fairness, protection of employees, participation.
Other stakeholders who support them	Employees, HSE manager, occupational health, HR; external union structures.
How they use the technology to achieve their goal	Do not use the system but are informed and consulted; can contest or negotiate the conditions of deployment.
Policies, regulations, social norms, technical constraints	Legal rules on informing and consulting employee representatives; collective agreements; data protection rules.

Stakeholder values (from the stakeholder × values matrix)	H: Safety (people, technical safety), Social well-being (workload, jobs), Fairness (diversity, non-discrimination), Confidentiality, Explainability, Accountability, Human control
Societal values extracted from the persona	Decent work, social justice, privacy, worker participation.

★ S11 – Company management	
Group / use cases	AI Application Maker – Facilitators / UC1, UC2
Primary goal	Improve competitiveness and working conditions through automation, within budget, while staying compliant and protecting the company's reputation.
Likes about their job and interaction with the company	Measurable productivity and cost gains; a modern and attractive workplace; fewer accident-related absences.
Dislikes about their job and interaction with the company	Cost and time overruns; conflicts with staff representatives; legal or reputational risk after an incident or data breach; dependence on a single supplier.
Company / group values	Competitiveness, sustainability, employer responsibility, reputation.
Other stakeholders who support them	Managers (logistics, production, maintenance, HSE), IT team, legal and data protection officer, staff representatives, development team.
How they use the technology to achieve their goal	Does not use the system; decides on investment, scope and conditions of deployment, and reviews reports and indicators.
Policies, regulations, social norms, technical constraints	Budget; labour law and the employer's safety obligations; data protection and AI regulation; corporate policies.
Stakeholder values (from the stakeholder × values matrix)	H: Economic sustainability (cost, productivity), Environmental well-being, Reliability, Safety (people, technical safety), Accountability, Regulatory compliance, Deployment & sovereignty
Societal values extracted from the persona	Economic sustainability, decent work, compliance and accountability, environmental sustainability.

2. List of values

Societal & individual values

Individual values impacted by the AI Application

Value	How the AI application affects it
Safety	People share space with moving robots and may take over machine tasks.
Human control	Being able to give, change, stop, refuse or correct a robot's action in time.
Confidentiality	Cameras, lidar and microphones capture images, voices and movements of people.
Ease of use	Voice-only or sound-only interaction can be hard or impossible for some people, or in a noisy workshop.
Reliability	Operators depend on robots working properly across everyday situations.
Fairness (non-discrimination)	Equal treatment between people and teams, including when resources are arbitrated and for vulnerable employees.
Social well-being	Workload, task takeovers, time pressure and job changes linked to automation.
Explainability	Understanding why a robot refuses, waits or prioritises a task.

Societal values impacted by the AI Application

Value	How the AI application affects it
Safety	Health and safety at work; prevention of accidents between humans and robots.
Accountability	Who answers for an error or accident involving an autonomous robot.
Fairness (non-discrimination)	Non-discrimination and inclusion of all employees, including vulnerable people.
Social well-being	Decent work and working conditions in an automated industry.
Confidentiality	Privacy and data protection at work; control of sensitive data.
Environmental well-being	Energy, electronics and resources used by a robot fleet.
Regulatory compliance	Respect of labour, safety, data protection and AI regulation.

Criteria that realise the values

Value	Realised by criteria
Accountability	Deployment and sovereignty
Human agency, human oversight (Human control)	Latency, Processing mode
Privacy, data governance (Confidentiality)	Processing mode, Confidentiality, Deployment and sovereignty
Technical robustness, safety	Robustness, Latency, Processing mode
Transparency (Explainability)	Explainability, Maintainability
Social well-being, environmental well-being	Energy
Economic criterion (to be confirmed with the teacher)	Total cost